

Md. Mahmudul Hasan

+8801715567401 | dearmahmud.bd@gmail.com | [LinkedIn](#) | [GitHub](#) | Savar, Dhaka, Bangladesh

SUMMARY

Full Stack Developer with hands-on experience building production-grade APIs and modern web applications using Node.js, Express.js, React.js, and Next.js. Passionate about clean architecture and systems design I build things from scratch to understand how they work at a deeper level.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++
Backend: FastAPI, Django, Express.js, Node.js, NestJS
Databases & Caching: PostgreSQL, MongoDB, Redis, Elasticsearch, SQLite
ORM & Messaging: SQLAlchemy, Alembic, Prisma, Mongoose, Motor, Apache Kafka
Auth & Security: JWT, bcrypt, RBAC, OAuth2
Infrastructure: Docker, Docker Compose, Nginx, GitHub Actions, Git, Vercel
Frontend: React.js, Next.js, Redux, HTML, CSS, SCSS
Core Concepts: REST API Design, Distributed Systems, Event-Driven Architecture, OOP, DSA

EXPERIENCE

Jr. Backend Developer

June 2025 – November 2025

Backbencher Studio

- Built and optimized RESTful APIs with Express.js and Node.js, implementing JWT-based authentication and resolving production issues to improve system reliability.
- Contributed to backend integrations for filtering, order management, AI integration and data pipelines.
- Collaborated with team using Git and GitHub for version control, code review, and feature branching.

PROJECTS

MahmudCore | *Python, WSGI, Jinja2, WebOb, PyPI* | [PyPI](#) | [GitHub](#)

- Built a WSGI-compliant Python web framework from scratch, implementing core internals including URL routing with dynamic parameters, class-based and function-based request handlers, and HTTP method control.
- Engineered a middleware pipeline architecture, Jinja2 template engine integration, custom response helpers (JSON/HTML/text), static file serving via WhiteNoise, and a global exception handling system.
- Published to PyPI with pytest coverage and semantic versioning, installable via `pip install MahmudCore`.

Redis-Inspired In-Memory Store | *Python, TCP, Socket, Event Loop* | [GitHub](#)

- Built a Redis-inspired in-memory key-value store from scratch in Python, implementing a single-threaded event loop with non-blocking I/O via `select()` to handle multiple concurrent clients without locks or race conditions.
- Implemented TTL-based key expiration using both lazy (on-access) and active (background probabilistic) cleanup strategies, along with support for core Redis commands including `SET`, `GET`, `EXPIRE`, `TTL`, and `KEYS`.
- Engineered dual persistence mechanisms - AOF and RDB - enabling crash-safe recovery and data durability.

PHARMA Web Application | *Redux, Next.js, TypeScript, Node.js, Express.js, MongoDB, Mongoose* | [GitHub](#)

- Built a full-stack medicine management system with Next.js, TypeScript, role-based access, and admin controls.
- Developed secure RESTful APIs with JWT authentication, role-based authorization, and full medicine CRUD operations.
- Implemented search, filtering, and order management features with optimized Mongoose models for seamless API integration.

EDUCATION

National Institute of Textile Engineering and Research

Bachelor of Science in Computer Science and Engineering — CGPA: 3.60/4.00

Dhaka-1350, Bangladesh

Feb. 2022 – Jun. 2026

CERTIFICATIONS AND ACHIEVEMENTS

Supervised Machine Learning: Regression and Classification — Coursera (Andrew Ng)

Next Level Web Development — Programming Hero

1st Place, Intra Research Fair Competition - 2026 — CRIR (NITER)

NextGen Hakathon - 2025 — Participant (IIUC)